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Roll No

EX-801-GS

B.E. VIII Semester

Examination, December 2020

Grading System (GS)

Computer Aided Electrical Machine Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain what is Computer Aided Design? Explain by a flow chart?
b) Develop a general algorithm for designing an electrical machine using non linear programming technique.
2. a) Describe the design of armature for a D.C machine. How the windings can be designed and placed in the field as well as armature?
b) How M.M.F distribution in D.C machines can be calculated?
3. a) What are mathematical programming methods? Explain.
b) Enumerate the advantages of computer aided design of electrical machines over conventional approach of design.
4. What do you understand by the phenomena of cogging and crawling in an induction motor. What steps will you suggest at the design stage to avoid this.

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5.
 - a) Discuss the calculation criterion for shape and number of stator slots. How the insulation is provided for these stator slots?
 - b) Write an algorithm for optimal design of three phase induction motor.
6.
 - a) Explain the different constraints and limitations of a synchronous machine design.
 - b) What are specifications of the motors supplied to the consumers? How these specifications helps in designing of motors?
7.
 - a) What are the factors which decide the selection of variables for optimal design of 3 phase alternator? Explain.
 - b) Draw a flow chart for optimal design of armature winding of 3ϕ alternative.
8. Write short notes (any two). 7×2=14
 - a) Difference between power transformer and distribution transformer.
 - b) Difference between 3Φ squirrel cage induction motor and slip ring induction motor.
 - c) Algorithms for optimal designs by example
 - d) Programming techniques (LP & NLP)
